MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC TEST – 06 (PROBABILITY)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 14: Probability** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 2 VSAQ questions (2 marks each). Section C has 2 SA questions (3 marks each).
3. Section D has 2 Long Answer HOTS questions (5 marks each).
4. Show clear steps in all probability derivations and calculations.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. A girl calculates that the probability of her winning the first prize in a lottery is 0.08. If 6000 tickets are sold, how many tickets did she buy? **[1]**

Options: (a) 480 (b) 48 (c) 240 (d) 600

2. Savita and Hamida are friends. What is the probability that both will have different birthdays (ignoring a leap year)? **[1]**

Options: (a) $\frac{364}{365}$ (b) $\frac{1}{365}$ (c) $\frac{1}{7}$ (d) $\frac{365}{366}$

3. A number is selected from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. The probability that it is a root of the quadratic equation $x^2 - 6x + 8 = 0$ is: **[1]**

Options: (a) $\frac{1}{4}$ (b) $\frac{1}{8}$ (c) $\frac{3}{8}$ (d) $\frac{1}{2}$

4. In a spinning arrow game of chance, the wheel is divided into 8 equal sectors numbered 1 to 8. The probability that the pointer stops at an odd number greater than 3 is: **[1]**

Options: (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{8}$ (d) $\frac{1}{8}$

5. **Assertion (A):** In a bag containing 6 red balls and some blue balls, if the probability of drawing a blue ball is twice that of a red ball, then the number of blue balls is 12.
Reason (R): Let number of blue balls be x. Then $\frac{x}{6+x} = 2 \cdot \frac{6}{6+x} \implies x = 12$. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

6. A bag contains lemon flavoured candies only. Malini takes out one candy without looking into the bag. What is the probability that she takes out: (i) an orange flavoured candy? (ii) a lemon flavoured candy? **[2]**

7. Two dice are thrown at the same time. Find the probability of getting: (i) a doublet (ii) a sum of 10. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

8. A box contains 100 red cards, 200 yellow cards, and 50 blue cards. If a card is drawn at random from the box, find the probability that it will be: (i) a blue card (ii) not a yellow card (iii) neither yellow nor blue. **[3]**

9. In a bag there are 44 identical cards with figures of circle or square on them. There are 24 circles, of which 9 are blue and 15 are green. There are 20 squares, of which 11 are blue and 9 are green. One card is drawn at random. Find the probability that it has: (i) a square (ii) a green circle (iii) a blue figure. **[3]**

SECTION D: LONG ANSWER / HOTS (5 Marks Each) [10 Marks]

10. An archery target consists of five concentric circular scoring rings marked Gold, Red, Blue, Black, and White from the center outwards. The diameter of the central Gold region is 21 cm and each of the other bands is 10.5 cm wide. An arrow is shot and hits the target at random. Taking $\pi = \frac{22}{7}$: (i) Find the area of the Gold region. (ii) Find the total area of the entire target (all 5 regions). (iii) Find the probability of hitting the Gold region. (iv) Find the area of the Red region and the probability of hitting the Red region. (v) Find the ratio of the probability of hitting the Gold region to that of hitting the White region. **[5]**

11. A jar contains red, white, and blue marbles. The probability of selecting a red marble is $\frac{1}{4}$ and that of selecting a white marble is $\frac{1}{3}$. The jar contains 10 blue marbles. (i) Find the probability of selecting a blue marble. (ii) Find the total number of marbles in the jar. (iii) Find the number of red marbles and white marbles in the jar. (iv) If 4 red marbles are removed and 2 blue marbles are added, what is the new probability of drawing a red marble? **[5]**

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